



**iRA
Synergy**
— Private Limited —
Nashik, Maharashtra, India

iRA's ECO
CLEANER • GREENER • SMARTER



SUSTAINABLE SOLUTIONS
FOR A BETTER TOMORROW

Containerized Bio-Gas Plant

**ENGINEERED FOR EFFICIENCY.
BUILT FOR SUSTAINABILITY.**

Transform Food Waste into
Nutrient-Rich Compost.
Cleaner Process. Greener. Future.



**SAFE &
DURABLE**

Built with high-quality
materials for long-
lasting performance.



**LOW
MAINTENANCE**

Designed for easy
operation and minimal
maintenance.



**ECO
FRIENDLY**

Converts organic
waste into clean and
renewable energy.



**COMPLETE
SOLUTION**

Integrated digester,
gas holder, and
control system.

CONTAINERISED
BIO-GAS PLANT MACHINE VIEW

**FRONT
VIEW**



**BACK
VIEW**



KEY FEATURES



**ECO
FRIENDLY**

Converts organic waste into clean and renewable energy.



**SAFE &
DURABLE**

Built with high-quality materials for long-lasting performance.



**LOW
MAINTENANCE**

Designed for easy operation and minimal maintenance.



**HIGH
EFFICIENCY**

Continuous & automatic operation for maximum output.



**COMPLETE
SOLUTION**

Built with high-quality materials for long-lasting performance.



**CLEAN
ENERGY**

Produces clean, smokeless biogas for multiple applications.

TECHNICAL SPECIFICATIONS

ABOUT THE PRODUCT

The Containerized Bio-Gas Plant is a compact and efficient waste-to-energy solution that converts organic waste into clean biogas and nutrient-rich organic fertilizer. Built within a durable shipping container, it offers quick installation, easy transportation, and minimal site preparation.

Using advanced anaerobic digestion technology, the system processes food, agricultural, and biodegradable waste to generate renewable energy while reducing waste disposal costs and environmental impact. Its automated operation, modular design, and reliable performance make it ideal for industries, farms, institutions, hotels, and municipal waste management projects.

Turn Organic Waste into Clean Energy with a Smart, Sustainable, and Scalable Solution.



TECHNICAL SPECIFICATIONS

PARAMETER	DETAILS
 Machine Type	Containerised Bio-Gas Plant
 Capacity	Customizable (as per requirement)
 Feedstock	Organic Waste
 Processing Time	24 - 36 Hours
 Power Requirement	Three Phase
 Body Material	Stainless Steel / Mild Steel (as required)

PARAMETER	DETAILS
 Heating System	Inbuilt Heater
 Digester Type	Anaerobic Digester
 Gas Storage	Integrated Gas Holder
 Mixing System	Automatic Agitaor
 Control Panel	HMI / PLC Based
 Installation	Plug & Play Installation



HIGH EFFICIENCY

Complete combustion with minimal residue.



LOW EMISSION

Environment friendly incineration with controlled emissions.



AUTOMATIC OPERATION

PLC based system for safe and hassle-free operation.



COST EFFECTIVE

Low fuel consumption and reduced operating costs.



BUILT TO LAST

Robust construction for long service life and reliable performance.

HOW IT WORKS



1 WASTE INPUT



Organic waste is collected and fed into the system through the inlet.



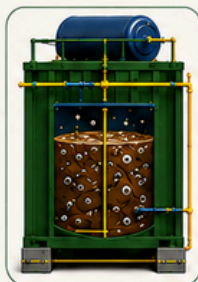
2 PRE-TREATMENT & MIXING



The waste is mixed with water to form a homogeneous slurry in the mixing tank.



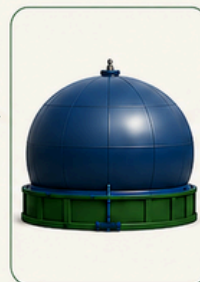
3 ANAEROBIC DIGESTION



Microorganisms break down the organic matter in the absence of oxygen to produce biogas.



4 GAS COLLECTION & STORAGE



The biogas generated rises and gets stored in the integrated gas holder.



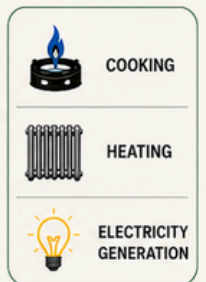
5 GAS PURIFICATION



The biogas is cleaned to remove impurities like H_2S , moisture and other particles.



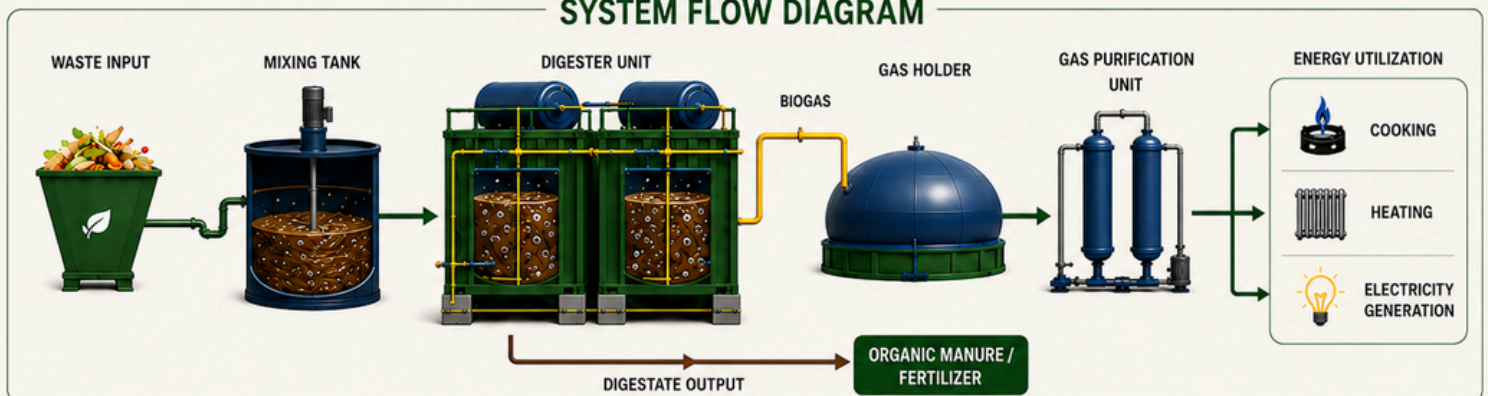
6 ENERGY UTILIZATION



Purified biogas is used for cooking, heating or electricity generation.



SYSTEM FLOW DIAGRAM



WASTEMENT machines are designed and manufactured to meet the highest safety standards, ensuring reliable performance with complete peace of mind.



**SAFE TODAY.
SUSTAINABLE TOMORROW.**

APPLICATION AREAS

 FOOD & BEVERAGE INDUSTRIES	 FARMS & AGRICULTURE	 SCHOOLS & UNIVERSITIES	 HOSPITALS & HEALTHCARES	 INSTITUTIONS & ORGANIZATIONS
 INDUSTRIAL ESTABLISHMENTS	 SHOPPING MALL & COMMERCIAL SPACES	 RESIDENTIAL SOCIETIES	 TEMPLES & RELIGIOUS PLACES	 PUBLIC TRANSPORT

IDEAL FOR THE DISPOSAL OF

 FRUIT & VEGETABLE WASTE	 COOKED FOOD WASTE	 EGG SHELLS	 TEA LEAVES & COFFEE GROUNDS
 BAKERY & BREAD WASTE	 FLOWER WASTE	 LEFTOVER MEALS	 CANTEEN & BUFFET WASTE

SAFETY FEATURES

WASTEMENT machines are engineered with intelligent safety systems to ensure operation, protect personnel, and deliver reliable performance every day.



OVERLOAD PROTECTION

Protects the systems from electrical overloads & short circuits for safe operation.



TEMPERATURE CONTROL

Automatic temperature monitoring and control maintain optimal digestion and prevent overheating.



VENTILATION SYSTEM

Proper ventilation ensures safe dispersion of gases and maintains a healthy working environment.



ALARM & AUTO ALERT SYSTEM

Instant alerts for any abnormal condition or system fault.



GAS LEAK DETECTION

Advanced gas sensors continuously monitor leakage and trigger alarms for immediate action.



PRESSURE SAFETY VALVES

Built-in pressure relief valves prevent overpressure & ensure safe gas handling.



EMERGENCY STOP SYSTEM

Easy access emergency stop button to quickly shut down the system in critical situations.



CONTINUOUS MONITORING

PLC based system continuously monitors and ensures safe run.

CERTIFICATES & COMPLIANCE

Manufactured in accordance with national and international standards to ensure quality, safety, and environmental responsibility.



WASTEMENT machines are designed and manufactured to meet the highest safety standards, ensuring reliable performance with complete peace of mind.



**SAFE TODAY.
SUSTAINABLE TOMORROW.**

SUSTAINABLE SOLUTIONS FOR A CLEANER TOMORROW

CLEANER ● GREENER ● SMARTER



ABOUT US

iRA Synergy Private Limited is committed to delivering innovative and sustainable waste management solutions that protect health, preserve the environment, and build a cleaner tomorrow.



OUR APPROACH



SUSTAINABLE BY DESIGN

Environmentally responsible solutions designed for long-term impact.



SAFE & RELIABLE

Engineered for safety, reliability and consistent performance.



ADVANCED TECHNOLOGY

Innovative technology for efficient and effective waste management.



COST EFFECTIVE

Low operating costs and minimal maintenance for long-term savings.



OUR COMMITMENT

We aim to create a healthier environment through responsible waste management, continuous innovation and a strong commitment to quality, safety and sustainability.



Clean Environment



Smart Solutions



Green Future

GET IN TOUCH



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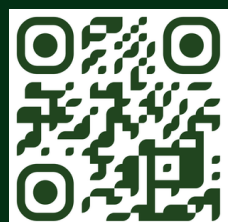


Nashik, Maharashtra, India

FOLLOW US



Let's work together for
a cleaner, greener future.



SCAN TO SEE
our solutions in action